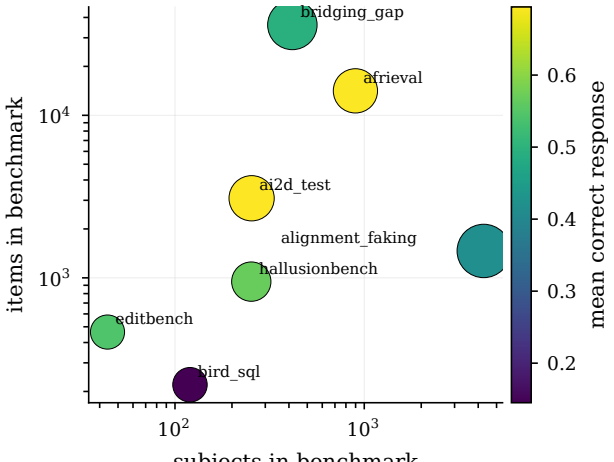
What is analysed in this appendix

Two different objects are kept separate: observed binary responses and frozen item-text embeddings.



Response graph

A bipartite edge joins a subject to an item only when that binary response was observed. Missing responses are not imputed.

Qwen item graph

Each filtered item connects to its 20 nearest 1024-d Qwen vectors by cosine similarity. It uses question text only.

Why separate them

Response structure needs labels and is training-only. Qwen edges can be built for a new item before its label is known.

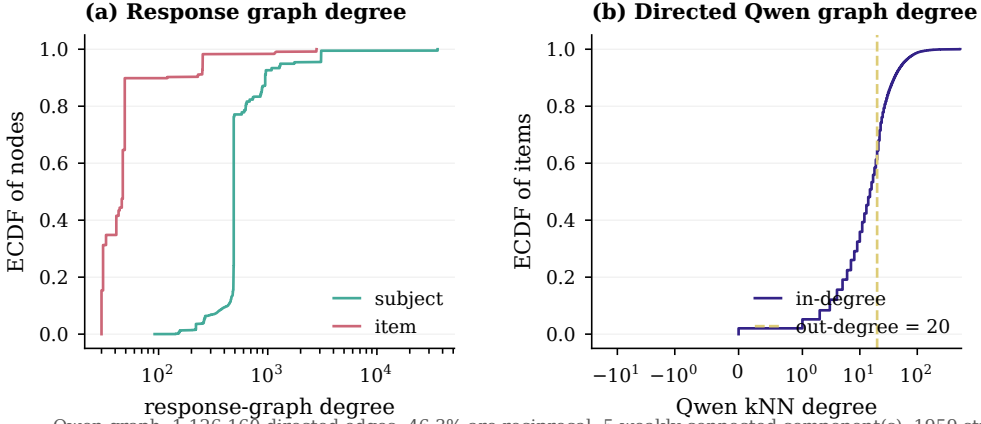
Filtering: 16,278,271 raw rows to 5,150,869; item support >= 30, subject-task responses >= 80.

How the two graphs are constructed

The response graph and the Qwen kNN graph have different edges, degree distributions, and test-time status.



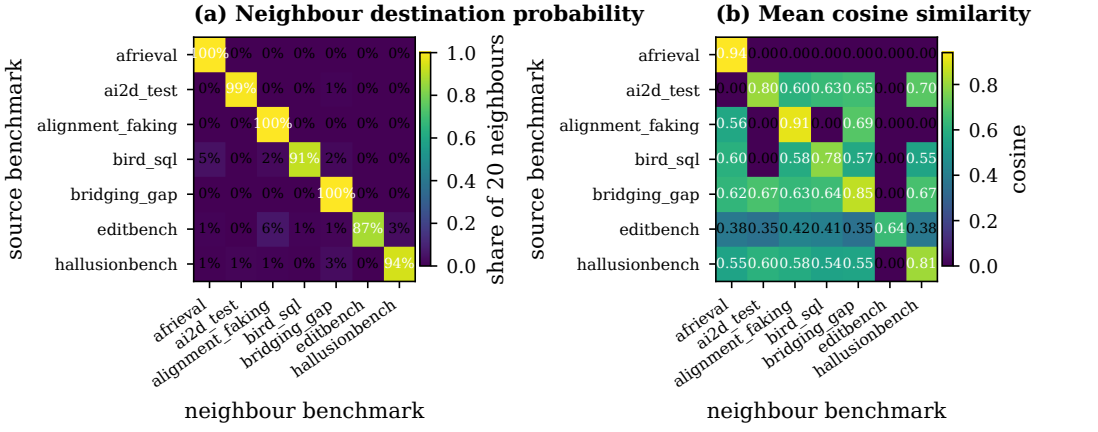
Direction comes from the source item's own Top-K list. It need not be mutual: $i \to j$ can exist while $j \not\to i$ does not.

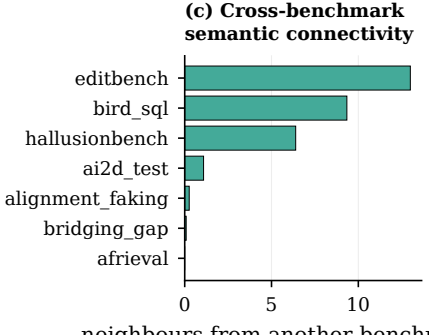


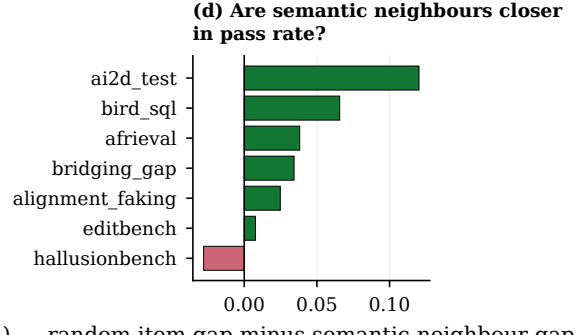
Qwen graph: 1,126,160 directed edges; 46.3% are reciprocal; 5 weakly connected component(s), 1959 strongly connected component(s).

Qwen3 item similarity forms a content-only graph

Each row contains the 20 nearest text-embedding neighbours of items from that benchmark. It is directed because each item chooses its own neighbours.



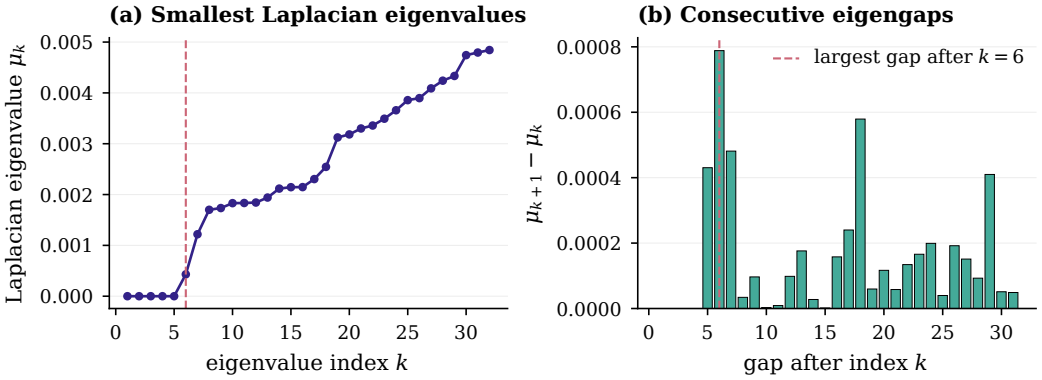




Panel (d): positive means Qwen neighbours have more similar observed pass rates than random items, after matching source and target benchmarks. It is descriptive, not a test-time score.

Laplacian spectrum of the symmetrised Qwen kNN graph

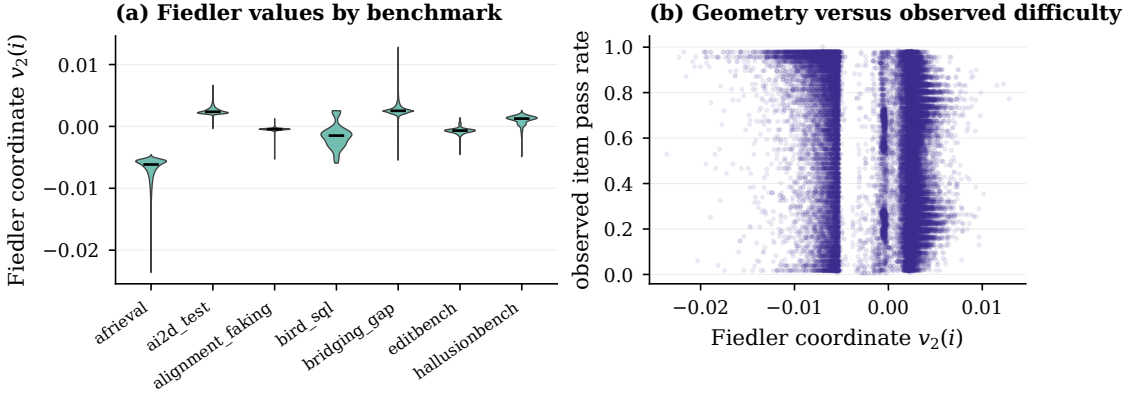
Cosine chooses each item's 20 neighbours; W is then the binary symmetrised kNN adjacency. $L_{\text{sym}} = I - D^{-1/2} W D^{-1/2}$. Near-zero values count disconnected pieces; a later eigengap suggests a coarse cluster count.



There are 5 weak components, so the first 5 eigenvalues should be near zero. The largest displayed gap is after k=6. This is a heuristic, not proof that k clusters are stable.

What the Fiedler vector can, and cannot, show

The Fiedler vector is the eigenvector for the second-smallest μ_2. It is computed inside the largest weak component because the global graph has several disconnected components and no unique global Fiedler vector.



Largest weak component: 56,221 items. The sign of v_2 is arbitrary. Separation by benchmark means the graph retains benchmark structure; a smooth relation to pass rate would suggest a difficulty-relevant semantic direction.